

product decays quickly.

This means two things:

- A feedback mechanism is a non-functional requirement. It must exist either inside the code or as a direct customer touchpoint.
- Developers need a top-down view—from business case to deployment—because AI is no longer an isolated IT delivery. It affects the business model, not just the software.

In earlier years, we did not revisit business cases frequently. In AI, we must. Needs evolve. Models drift. Use cases change direction. Developers can no longer focus solely on implementation; we need to understand the charter, the model, and the end-to-end refinement cycle.

Scale

The third principle is about scaling. Traditionally, we designed modules based on functions: Module 1 performs function A; module 2 performs function B. Testing followed the same pattern.

But AI development today requires a shift from functional thinking to service-level development. Why? Because every module now calls external services—LLMs, APIs, cloud platforms—not once, but many times in a single workflow. We no longer live in a world where only one or two modules connect externally. Now, every module does.

This is where microservices become essential. If your organisation uses GPT today and switches to another provider next year, how will your system adapt? If a model upgrades, will you redo your entire codebase?

With microservices, a change in one service can flow across the organisation without rewriting everything. This is the power of thinking in terms of services rather than functions.

Cloud providers have also evolved. Earlier, we worked with infrastructure-as-a-service. Today, we rely heavily on the platform-as-a-service model --

APIs, ML services, and LLM endpoints. So our architecture must match the service mindset.

Grow

The fourth principle is growth. To move forward responsibly, we must understand where we stand. Gartner's AI maturity model describes five levels:

1. **Awareness:** Employees understand AI basics.
2. **Active:** Teams experiment with POCs, hackathons and low-hanging use cases.
3. **Operational:** AI improves efficiency through internal operations. This is true for most organisations today.
4. **Product:** AI powers products delivered to customers with accuracy and reliability.
5. **Sentinel:** Fully autonomous, zero-human-touch decision-making systems. Autonomous driving is an example.

Importantly, organisations do not move through these levels sequentially. You cannot finish Level 1 and then begin Level 2. Instead, activities across all levels must run in parallel. For instance, a company can run an operational AI system while parallelly creating awareness programmes for responsible AI. Growth is not linear; it is layered.

- Awareness increases with company-wide training programmes, expert lectures, and AI meets.
- Active exploration comes from hackathons and quick POCs.
- Operational impact usually comes through central teams or centres of excellence that identify use cases and implement them.
- Product-level confidence comes with stronger computing, streamlined pipelines and model governance.
- Sentinel level is the long-term vision and represents the highest form of autonomous intelligence.

So why is AI suddenly at an inflection point? We have had forms of AI since the early 1900s. The term itself appeared in 1842 and was formally

coined in 1956. We have seen expert systems, rule-based systems, games such as Pac-Man and chess, and early healthcare applications. But the real acceleration began around 2011 with recommendation systems and the rise of machine learning and deep learning.

Today's inflection point is driven by four clear trends.

Data explosion: We collect data, knowingly and unknowingly, from mobile phones, vehicles, applications, and sensors. We have sufficient data to experiment, and algorithms can now generate synthetic data as well.

Cloud computing: Earlier, computing meant connecting to a mainframe somewhere else, often with delays and limitations. Today, a few clicks provide 4GB, 8GB, or 32GB of GPU power instantly. Cloud platforms have made high-end computing accessible to every developer.

Improved algorithms: New AI models appear constantly. But the real reason for this speed is that everything is open source. This is no longer enterprise controlled. Developers everywhere contribute to global progress.

Open contribution: The open source ecosystem is what accelerates this industry. Every improvement becomes available to everyone. This is why the field feels like it is moving at lightning speed.

AI is everywhere. AI is everything. But unless we define our own box, we will be overwhelmed.

No organisation progresses linearly, and no developer can afford to work without understanding the business and refinement cycle. AI demands a different way of thinking — structured, strategic and continuously evolving. 

This article is based on the session titled 'Beyond Automation - Strategic AI Integration with GenAI and LLM' by Pradeeba P., delivery manager, Thoughtworks, at AI DevCon in Bengaluru. It has been transcribed and curated by Apurba Sen, senior journalist at the EFY Group.